

# CARINGBAH HIGH SCHOOL



## YEAR 12 (2010) ASSESSMENT TASK 1 MATHEMATICS (2 UNIT) DECEMBER 2009

**Time Allowed:** 65 minutes

- Instructions:**
- \*Start each "Question" on a new page
  - \*Show all necessary working
  - \*Do not use liquid paper
  - \*Silent, nonprogrammable calculators may be used
  - \*Attempt all questions

**Marks**

### Question 1 (8 marks) (Start a New Page)

- a) The  $n$ th term of a sequence was given by  $T_n = 5 \times 2^{n-1}$ . Find  $T_3$ . (1)
- b) Differentiate with respect to  $x$ :
- (i)  $\sqrt{x}$  (1)
  - (ii)  $(4x-1)^6$  (1)
  - (iii)  $\frac{2x+1}{3x-1}$  (2)
- c) Find the value of ' $k$ ' if  $x^2 + 3x + k = 0$  has real roots. (3)

**Question 2 (7marks) (Start a New Page)****Marks**

a) Solve the inequality  $(x-2)(x+5) \geq 0$  (1)

b) Solve for  $x$ :  $x^4 - 3x^2 - 54 = 0$  (3)

c) Sketch an arc of function  $y = f(x)$  on a set of coordinate axes, given that: (2)

$$f'(x) > 0$$

$$f''(x) < 0$$

d) Using the information in the table below, what type of point is found at  $x=3$  for this function? (1)

|                 |      |   |      |
|-----------------|------|---|------|
| $x$             | 2.9  | 3 | 3.1  |
| $\frac{dy}{dx}$ | -0.8 | 0 | -0.6 |

**Question 3 (7 marks) (Start a New Page)**

a) For what value of  $k$  does the equation  $x^2 + kx - 8 = 0$  have a root of 3? (2)

b) If  $\alpha$  and  $\beta$  are the roots of the equation  $2x^2 + 5x - 1 = 0$  find, without solving the equation, the value of:

(i)  $\alpha + \beta$  (1)

(ii)  $\alpha\beta$  (1)

(iii)  $\alpha^2 + \beta^2$  (1)

c) If  $y = 5x^{3.4} - 3x$ , find  $\frac{d^2y}{dx^2}$  in simplest form. (2)

**Question 4 (8 marks) (Start a New Page)**

- a) If  $2x^2 + 3x + 1 \equiv A(x-1)^2 + B(x-1) + C$  find the values of  $A$ ,  $B$  and  $C$ . (3)
- b) Evaluate  $\sum_{r=1}^3 (r+5)$  (2)
- c) For the arithmetic sequence  $\{ 5, 8, 11, \dots \}$
- (i) find the general term  $T_n$  in its simplest form (2)
  - (ii) find which term is the first to exceed 200 (1)

**Question 5 (8 marks) (Start a New Page)**

For the curve with equation  $y = x^3 - 9x^2 + 24x + 2$

- (i) Show that there are stationary points at  $x = 2$  and  $x = 4$ . (2)
- (ii) Find the inflexion point (2)
- (iii) Sketch the curve, including the above information. (4)  
Use a half page diagram, correctly labelled. Include the  $y$ -intercept also.

**Question 6 (7 marks) (Start a New Page)****Marks**

- a) The cost of individual organ pipes of a church organ were in arithmetic progression. The smallest pipe cost \$32; the second pipe cost \$37; the third pipe cost \$42 and so on.
- (i) Find the cost of the  $k$ th pipe (2)
- (ii) If the **total** cost of the organ pipes was \$10443, find the number of organ pipes. (3)
- b) For what values of  $x$  is the function  $f(x) = 12x - x^2$  decreasing? (2)

**Question 7 (8 marks) (Start a New Page)**

- a) Given that the equation (2)
- $$3x^2 + (k-1)x + (k+4) = 0$$
- has roots that are reciprocals of each other, find the value of  $k$ .
- b) (i) Show why the geometric series (1)
- $$18 + 14.4 + 11.52 + \dots$$
- has a limiting sum.
- (ii) Find the limiting sum of the above series. (2)
- c) The line  $y = mx$  is a tangent to  $y = x^2 - 2x + 1$ . Find the (3)
- possible values of  $m$ .

**END OF EXAM**

Q1 a)  $T_3 = 5 \times 2^{3-1}$   
 $= 20$

b) i)  $\frac{d}{dx}(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}}$

ii)  $\frac{d}{dx}(4x-1)^6 = 6(4x-1)^5 \cdot 4$   
 $= 24(4x-1)^5$

iii)  $u = 2x+1$     $v = 3x-1$   
 $u' = 2$     $v' = 3$

$\frac{d}{dx} = \frac{(3x-1) \cdot 2 - (2x+1) \cdot 3}{(3x-1)^2}$

$= \frac{6x-2-6x-3}{(3x-1)^2}$

$= \frac{-5}{(3x-1)^2}$

c)  $\Delta = (3)^2 - 4(1)(k)$   
 $= 9 - 4k$

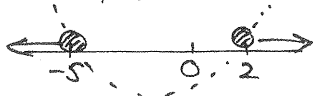
Let  $\Delta \geq 0$  for real roots

$9 - 4k \geq 0$

$9 \geq 4k$

$k \leq 2\frac{1}{4}$

Q2. a) b'dary pts  $x = 2, -5$



$x \leq -5, x \geq 2$

b) Let  $v = x^2$

$v^2 - 3v - 54 = 0$

$(v-9)(v+6) = 0$

$v = 9, -6$

$\therefore x^2 = 9$  only

$x = \pm 3$

c)  $y = f(x)$

d) A horizontal inflexion.

Q3 a)  $(3)^2 + k(3) - 8 = 0$

$3k + 1 = 0$

$k = -\frac{1}{3}$

b) i)  $\alpha + \beta = \frac{-(5)}{(2)} = -2\frac{1}{2}$

ii)  $\alpha\beta = \frac{(-1)}{(2)} = -\frac{1}{2}$

iii)  $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2(\alpha\beta)$   
 $= (-2\frac{1}{2})^2 - 2(-\frac{1}{2})$

$= 7\frac{1}{4}$

c)  $\frac{dy}{dx} = 17x^{2.4} - 3$

$\frac{d^2y}{dx^2} = 40.8x^{1.4}$

Q4. a)  $A(x-1)^2 + B(x-1) + C$

$= Ax^2 - 2Ax + A + Bx - B + C$

$= Ax^2 + (-2A + B)x + (A - B + C)$

$= 2x^2 + 3x + 1$

$\therefore A = 2$

$-4 + B = 3$

$\therefore B = 7$

$2 - 7 + C = 1$

$\therefore C = 6$

b)  $(1+5) + (2+5) + (3+5)$

$= 6 + 7 + 8$

$= 21$

c) i)  $a = 5$   $d = 3$

$$T_n = 5 + (n-1)3$$

$$= 3n + 2$$

ii) let  $T_n > 200$

$$3n + 2 > 200$$

$$3n > 198$$

$$n > 66$$

ie. the 67th term.

Q5.  $y = x^3 - 9x^2 + 24x + 2$

i)  $\frac{dy}{dx} = 3x^2 - 18x + 24$   
 $= 0$  for st.pts.

$$3(x^2 - 6x + 8) = 0$$

$$3(x-2)(x-4) = 0$$

$$\therefore x = 2, 4$$

ii)  $\frac{d^2y}{dx^2} = 6x - 18$   
 $= 0$  for a poss. inflex?

$$\therefore x = 3$$

$$x = 3 \rightarrow y = (3)^3 - 9(3)^2 + 24(3) + 2$$

$$= 20$$

The inflexion is at  $(3, 20)$

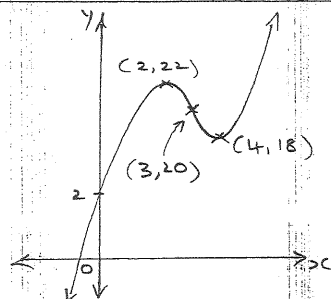
(note that because the curve is a cubic poly., and therefore continuous, there must be an inflexion)

iii)

$$x = 2 \rightarrow y = 22$$

$$x = 4 \rightarrow y = 18$$

y-intercept (when  $x=0$ ) is  $y = 2$ .



Q6 a) i)  $\{32, 37, 42, \dots\}$

is an A.P. with  $a = 32$ ,  $d = 5$

$$T_n = 32 + (n-1)5$$

$$= 5n + 27$$

$$T_k = 5k + 27 \text{ (\$)}$$

ii)  $S_n = \frac{n}{2}(a + l)$

$$\frac{n}{2}(32 + 5n + 27) = 10443$$

$$n(5n + 59) = 20886$$

$$5n^2 + 59n - 20886 = 0$$

$$n = \frac{-59 \pm \sqrt{59^2 - 4 \times 5 \times -20886}}{10}$$

$$= \frac{-59 \pm 649}{10}$$

$$= 59 \text{ only } (n > 0)$$

Hence there are 59 pipes

b)  $f'(x) = 12 - 2x$

Let  $f'(x) < 0$  to be decreasing

$$12 - 2x < 0$$

$$12 < 2x$$

$$x > 6$$

Q7 a)  $\alpha\beta = 1$

$$\therefore \frac{k+4}{3} = 1$$

$$k = -1$$

b) i)  $r = \frac{14.4}{18} = 0.8$

Hence  $|r| < 1$

$\therefore S_{\infty}$  exists

ii)  $S_{\infty} = \frac{18}{1-0.8}$

$$= 90$$

c)  $\begin{cases} y = mx & \text{--- ①} \\ y = x^2 - 2x + 1 & \text{--- ②} \end{cases}$

Let ① = ②

$$x^2 - 2x + 1 = mx$$

$$x^2 + (-2-m)x + 1 = 0$$

$$\Delta = (-2-m)^2 - 4(1)(1)$$

$$= 4 + 4m + m^2 - 4$$

$$= 4m + m^2$$

Let  $\Delta = 0$  for tangency

$$\therefore m(m+4) = 0$$

$$\therefore m = 0, -4.$$